EX.No.2.1 CREATE A 3D SOLID MODEL FROM ISOMETRIC DRAWING

AIM:

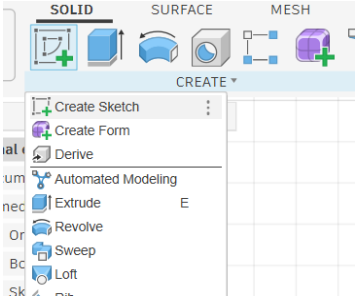
To create a 3D model from the given isometric drawing using fusion 360 tool.

REQUIREMENTS:

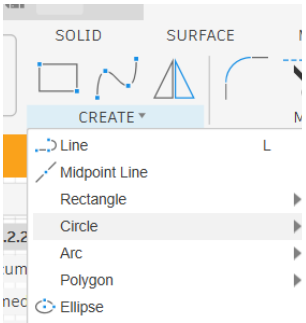
1. Autodesk fusion 360
2. Windows 10
3. 16 GB of RAM (integrated graphics recommend 1 GB or more)
4. 2.5 Mbps or faster download; 500 Kbps or faster upload.

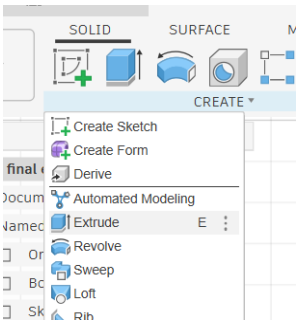
PROCEDURE:

1.Create the base
Open Fusion 360, select Create Sketch, choose the XY plane, draw a rectangle of mm, and finish the sketch. Use Extrude to give it a height of 100×50 mm.



2.Create the front cylinder
Select the front face and create a sketch. Draw a circle of 50 mm diameter, then use Extrude to extend it outward by the length shown in the drawing.

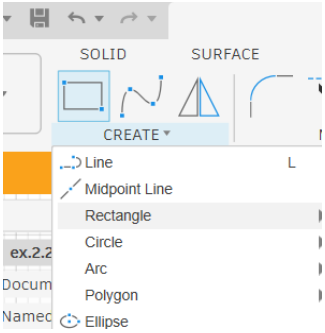


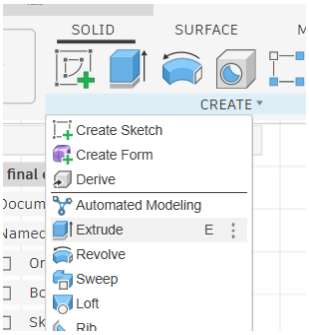


3. Create the sloping top feature
Select the top face and sketch the raised portion using the given dimensions. Finish the sketch and use Extrude to raise it.Create the inclined face according to the drawing.

4. Create the top circular boss
Sketch a circle of 20 mm diameter on the right side of the top face. Finish the sketch and extrude it upward to the required height.

5. Create the small square feature
Sketch a small square on the left side of the top face and extrude it upward using the dimensions given in the drawing.





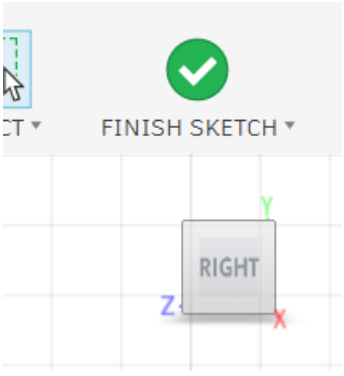
6. Finish the model
Check the dimensions, adjust the positions of all features, and select Finish Sketch whenever required. Save the completed model.Create a line

*Select Line command.

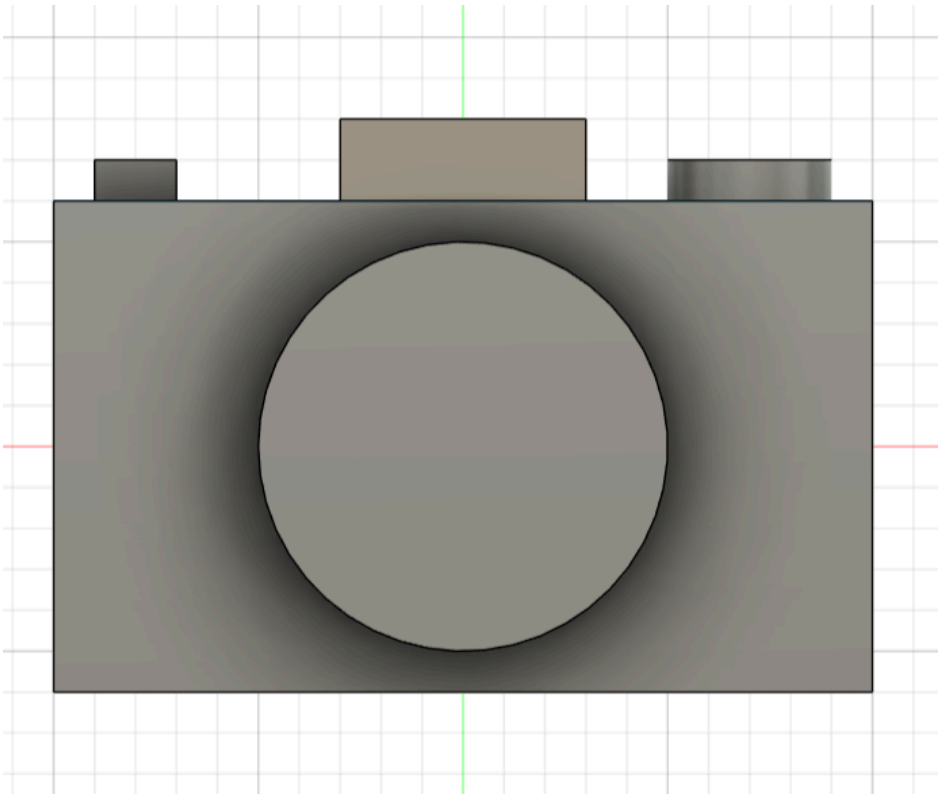
* Draw the projections in the diagram

* Click Finish Sketch.

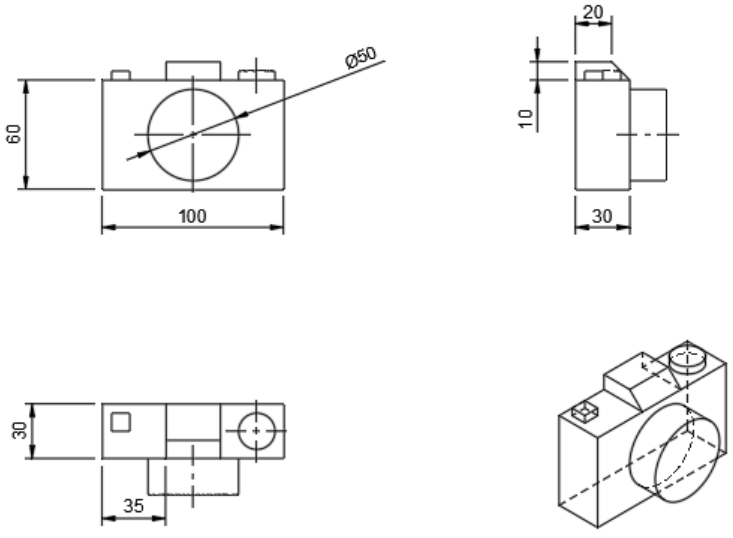
* extrude the sketch.



INPUT:



OUTPUT:



RESULT:

Thus, the 3D model is created from the given isometric drawing.